Interference Protection of Radio Astronomy Services Using Cognitive Radio Spectrum Sharing Models

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Abstract: - Radio astronomical services (RASs) use the radio spectrum to observe cosmic sources (Srcs). They are protected from interference by establishing national radio quiet zones (NRQZs). However, NRQZs do not prevent interference from satellite networks with intersatellite links. This causes impairment in directly observing cosmic sources. Therefore, current knowledge of the universe is inaccurate being obtained from filtered signals. However, accurate knowledge can be obtained in RASs via a pristine observation of Srcs using cognitive radio (CR) technology. This paper, examines the suitability of CR spectrum sharing models for interference protection of RASs. It proposes an interference protection framework and investigates the satellite's transmit opportunities using visibility duration data of RASs. It then examines the end to end delay of the satellite network when the framework is used. Simulation is conducted for a constellation of low earth orbit satellites and the obtained results show that transmit opportunities exist for intersatellite links when the interweave spectrum sharing model is used without causing any interference to RASs or significantly increasing the end to end delay.

Keywords: Kepler's Laws; Cognitive radio; Radio astronomy; Spectrum sharing

I. INTRODUCTION

Currently, there is an increased interest in radio astronomy as seen in the number of radio astronomical observatory (RAO) projects. On the other hand, there is also an increased proliferation of wireless networks with higher demand for bandwidth by subscribers for multimedia applications. These increased spectrum demand by these services results in a clash between both radio astronomy and wireless networks causing interference.

The interference protection of radio astronomical services (RASs) is realized by establishing national radio quiet zones (NRQZs). However, NRQZs are insufficient for protecting RASs from interfering satellite networks. The effects of interfering satellite signals on RASs data interpretation are reduced using signal processing techniques that flag and remove interferer's signals prior to data interpretation.

Moreover, the advent of satellites with adaptive transmit waveforms limit the performance of signal processing algorithms. Interference also results when a satellite that has known transmit waveform is de-orbited and replaced by another with an unknown transmit waveform. The NRQZ and signal processing techniques are not suitable to handle the resulting interference in these scenarios. These limitations can be addressed using cognitive radio technology (CRT). The use of CRT also enables an improved knowledge of the universe to be obtained because it enables improved observations of astronomical sources.

Cognitive radio networks (CRNs) are next generation wireless network standards that incorporate the feature of dynamic spectrum access (DSA). CRNs comprise primary users (PUs) and secondary users (SUs). PUs have a higher spectrum access priority than SUs. The CRN protocol stipulates that SUs shouldn't interfere with PUs.

DSA enables CRNs to avoid interfering with existing services such as the TV broadcasting service [1], terrestrial and satellite networks [2]. These services use the 0.1-6 GHz spectrum that is also of interest to the RAS. Some bands in this range are used for intersatellite links (ISLs) in satellites networks [5]. These ISLs interfere with RASs.

Bentam *et al.* [6, 7] consider the use of CRN techniques to prevent interference between RAS and wireless networks. They conclude that it is impossible for RASs and wireless networks to share spectrum after performing interference level studies. However, their examination of the interference prevention coexistence feasibility is based on the underlay spectrum sharing model. They have not considered the overlay and interweave spectrum sharing models.

This paper extends Bentam *et al.* [6, 7] and explores the feasibility of using the overlay and interweave spectrum sharing models [8] to design mechanisms for protecting RASs from interfering ISLs. It makes the following contributions:

1. First, it examines the design tradeoffs when CRN spectrum sharing models are used for the interference protection of the RAS.
2. Second, it presents a framework that can be used to realize interference protection when CRN spectrum sharing techniques are used.
3. Third, it examines the transmission opportunities for intersatellite links based on the spectrum usage pattern of the RAS.
4. Fourth, it investigates the connection duration available to SUs and the end to end delay when the proposed interference protection framework is used.

The rest of the paper is organized as follows. Section II discusses related literature. Section III presents the problem. Section IV describes tradeoffs of different CRN spectrum sharing models. Section V focuses on the incorporation of cognitive radio in RAS. Section VI describes simulation results. Section VII concludes the paper.

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II. RELATED LITERATURE

Advances in RAS have been made in the areas of telescope design from single antennas, interferometers, very long baseline interferometers [9]; to giant metrewave radio telescopes, very large array telescope and the square kilometer array (SKA) with improved sensitivity and resolution [10].

Pankonin *et al.* [9] propose granting of primary spectrum to the RAS to improve the interference protection capability of NRQZs. However, a mechanism is needed to inform communication satellites of this primary access.

Waterman [11] recognizes that ISLs using the 380-400 MHz spectrum reduces the reliability of astronomical observations. An additional mechanism is required to ensure that ISLs using this band do not reduce the reliability of astronomical observations.

Jonas [12] improves upon [11] by identifying the 1.63-1.73GHz and 2.20-2.40 GHz spectra as additional bands suitable for RAS and wireless communications. The Iridium constellation and the GLONASS network are also recognized to interfere with the RAS. However, a mechanism is required to protect RASs from interfering Iridium and GLONASS satellites.

Gergely [13] notes that NRQZs are not globally adopted to protect RAS from interference. The paper proposes that satellite motion should be restricted over the telescope's vicinity. However, a suitable mechanism is required to ensure that this restriction does not affect the end to end delay of satellite networks and further analysis is required.

Gergely [14] outlines the progression in the reservation of the 1.61-1.63 GHz and 71-275 GHz bands for the RAS and advances [13] by considering interfering effect of services on harmonically related bands. However, mechanisms are required to prevent interference between RASs and ISLs using harmonically related bands when satellite motion in the Iridium network and ISLs activation is not restricted.

DeBoer *et al*. [15] focus on the use of spectrum allocation policies to prevent interference between RASs and wireless communication networks by reserving some channels for radio astronomy. However, this method does not consider the scarcity of the radio spectrum resource and limits the achievable spectrum utilization.

Gergely [16] extends [12-13] and note that the increased interest in astronomy requires bands spanning 0.1 - 6GHz. The paper canvasses for a greater awareness of the spectrum demands for RASs in spectrum regulations at global and national levels. However, the proposal of interference protection mechanisms that improve spectrum utilization using CRN spectrum sharing techniques has not been considered.

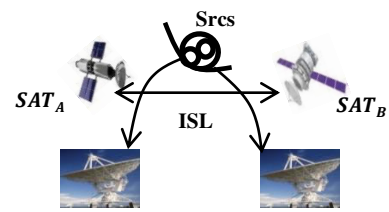
Ford *et al.* [17] examines technical and non-technical methods used for interference protection of RASs. Some of the technical and non-technical methods used are offline post data processing and NRQZs respectively.The sustainability of using NRQZs is limited because of factors like, rapid urbanization, industrialization and mineral exploration motives. Therefore, new mechanisms are required to protect the RAS from interference considering these limitations.

III. PROBLEM DEFINITION

The scenario considered in this paper comprises cosmic sources (Srcs) being observed by the RAO, satellite A, SAT_A and satellite B, SAT_B in the RAO vicinity. SAT_A and SAT_B are linked via ISLs. The Srcs are the PUs while the satellites linked with ISLs are the SUs.

The interaction between SUs and PUs is shown in Figure 1. Interference occurs when the observation of the Srcs and data forwarding via the ISLs simultaneously occur on the same path.

Let $I_X \in \{0,1\}$ where $X \in \{Srcs, ISL\}$ be an indicator variable such that $I_X = 0$ and $I_X = 1$ indicates the absence and presence of X respectively. Interference results when $I_{X=Srcs} = 1, I_{X=ISL} = 1$.



$$I_{X=Src} = 1, I_{X=ISL} = 1$$

Figure 1: Interference between ISLs and the Srcs.

IV. SPECTRUM SHARING MODELS

This section examines the tradeoffs when the overlay and interweave CRN spectrum sharing models are used for interference protection of RASs.

A. Overlay Spectrum Sharing Model

In this model, satellites are expected to be capable of transmitting own and astronomical data. The existing ground satellite infrastructure receives astronomical data from space based astronomy systems and forwards it to the astronomical data processing terminal.

This model does not degrade the SU Shannon channel capacity as the transmit power is not reduced. The end to end delay of forwarding data through the satellite network is also unaffected. However, because of the need to process and forward astronomical data, the use of this model increases the computational demand on satellites. Therefore, it is important that satellites have additional computational capabilities. This necessitates the replacement of monolithic satellites or adding new modules to fractionated satellites. Hence, this model has the demerit of having a high implementation cost.

B. Interweave Spectrum Sharing Model

This model permits SUs to transmit when there are no emissions from Srcs (i.e no PUs). Srcs are natural satellites whose visibility is governed by Kepler's laws. The transmit power of SUs are not limited and mechanisms using visibility epoch and duration of Srcs are required to enable interfering satellites to determine the epochs and duration of ISL activation.

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The deactivation of ISLs increases the end to end delay when the shortest path in the satellite network lies in the NRQZ vicinity. Therefore, new mechanisms are needed to avoid an increased end to end delay when this sharing model is used.

This model does not require the replacement of satellites or significant additional computational capabilities since satellites do not process astronomical data. Moreover, the model also enables pristine observation of Srcs, thereby improving the knowledge of the universe.

C. Proposed Framework

The framework is based on the interweave spectrum sharing model because of the low implementation cost and knowledge improvement potential. It comprises the observation plan (OBL), astronomical spectrum database (ASD), cognitive reasoner (CRE) and cognitive ISL activator (CSLA). It prevents interference by ensuring that $I_{X=Srcs} = 1, I_{X=ISL} = 0$ or $I_{X=ISL} = 1,\ I_{X=Srcs} = 0$.

The OBL is a micro-description of Srcs observed daily at a frequency used by ISLs. The ASD is derived by aggregating several OBLs. The OBL and ASD are located at the RAO while the CRE and CSLA are on the satellite.

The RAO sends the ASD to the CRE via the plan acquisition channel (PAC). The CRE determines when a satellite is in the observatory vicinity and uses its awareness of the satellite network constellation to determine the interfering satellites. The receiving satellite transmits a beacon to satellites in the RAO vicinity. The CSLA on affected satellites deactivates their ISLs. The interaction between these entities is shown in Figure 2.

The ISL has a link budget described by transmit antenna gain of satellite,i, $G_A(i)$, transmit antenna gain of satellite,j, $G_A(j)$, satellite i processing gain of satellite, $G_P(i)$, satellite j processing gain $G_P(j)$, antenna mis-pointing loss between satellites i and j, $M_p(i,j)$, propagation loss between satellites i and j, $P_i(i,j)$ and free space loss $FSPL_{ISL}(i,j)$. The ISL's fade margin $FM_{ISL}(i,j)$ can be computed as:

$$FM_{ISL}(i,j) = P_t(i) + G_A(i) + G_A(j) + G_P(i) + G_P(j) - (M_p(i,j) + P_i(i,j) + FSPL_{ISL}(i,j)) \quad (1)$$

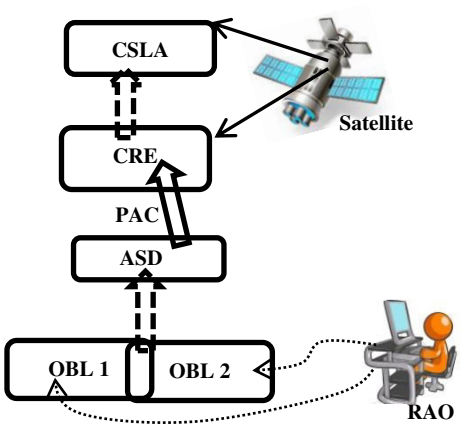

Figure 2: Interaction between entities in the proposed framework

The interweave spectrum sharing model does not vary any of the variables in (1). Therefore, the framework does not affect the ISL Shannon channel capacity, $C_{ISL}(i,j)$. However, the end to end delay is affected because of change in number of en-route satellites when ISLs are activated or deactivated.

In analyzing the end to end delay, we consider a case where the satellite network comprises a total M low earth orbit satellites with N satellites in the NRQZ vicinity. Given that $L_{ISL}(i,j)$ bytes are sent via the ISL between satellites i and j, and $N_{ISL}(i,j)$ is the number of ISLs between satellites i and j. The end to end delay in the absence of the framework $D_{N|F}(i, j = i+1)$ is:

$$D_{N|F}(i,j) = \sum_{i=1}^{M-1} I_{ISL}(i,j)((N_{ISL}(i,j))\frac{L_{ISL}(i,j)}{C_{ISL}(i,j)} \quad (2)$$
$$j = i+1$$

$I_{ISL}(i,j)$ is the ISL indicator variable between satellites i and j. $I_{ISL}(i,j) \epsilon \{0,1\}$ such that $I_{ISL}(i,j) = 0$ and $I_{ISL}(i,j) = 1$ signifies ISL deactivation and activation respectively. The end to end delay in (2) is that of the network in [18] without the framework. The end to end delay $D_F(i,j)$ when the framework is used is:

$$D_F(i,j) = \sum_{i=1}^{M-(N)} \sum_{j=1}^{(N)} I_{ISL}(i,j)(N_{ISL}(i,j))\frac{L_{ISL}(i,j)}{C_{ISL}(i,j)} \quad (3)$$

We also examine how interorbital links (IOLs) between satellites in the low earth and higher altitude orbits affect the end to end delay in the framework. Higher altitude satellites have their beams directed outside the observatory vicinity. The end to end delay when the network comprises n higher altitude orbits satellites,$D_{F,1}(i,j)$ is:

$$D_{F,1}(i,j) = \sum_{i=1}^{M-(N+n1)} \sum_{j=1}^{n} A(i,j') + B(i,j) \quad (4)$$

Where:

$$A(i,j') = I(i,j')(N_{ISL}(i,j'))\frac{L_{ISL}(i,j')}{C_{ISL}(i,j')}\ ;\ j' = i+1$$
$$B(i,j) = I(i,j)(n(i,j))\frac{L_{ISL}(i,j')}{C_{ISL}(i,j')}$$

V. SIMULATION AND DISCUSSION

This section presents results on the investigation of the transmit opportunities and the end to end delay.

A. Investigation of Transmit Opportunities

The ISLs's transmission opportunities are investigated using the KAT-7 telescope observation data for an observation bandwidth of 210MHz. The data spans the following periods: period 1 (06/11/2012), period 2 (21/01/2013) and period 3 (16/02/2013). These periods have the duration 17:15:54.8 - 23:37:42.6, 16:13:50.4-20:17:30.6 and 08:42:05.3-16:40:55.7 respectively.

Data preprocessing is done using CASA version 3.4.0 and further investigation of transmit opportunities is done via MATLAB. Although, visibility data for telescopes operating at the higher Gigahertz spectrum bands are desirable, these data are currently unavailable. However, the results presented here demonstrates the feasibility of using Kepler's laws as a basis of using the interweave

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spectrum sharing model for the design of the proposed framework.

The spectrum bands for the obtained data lie in 1.80-1.95 GHz. The transmit opportunities are investigated to obtain the spectrum usage of RASs and the back to back transmit opportunities for ISLs.

Analysis of the spectrum usage is done by investigating spectrum utilization of RASs and the ISL's transmit opportunities. The result of this investigation is shown in Figure 3. Results show that the daily spectrum usage of RASs is low with existing transmit opportunities for ISLs in the 1.8-1.95 GHz band for the considered periods.

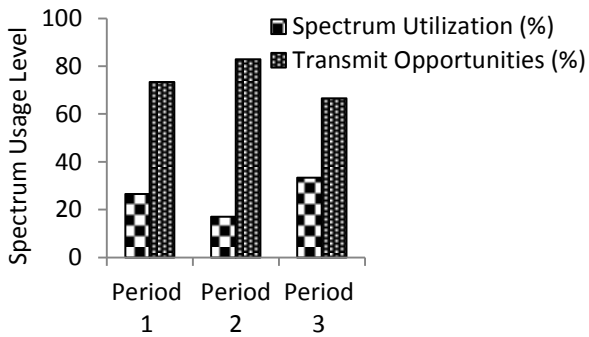

Figure 3: Spectrum utilization and transmit opportunities.

The back to back transmit opportunities available to the ISLs is also investigated. This enables additional connection duration of ISLs between the observations of different Srcs. Period 3 is considered in this analysis because the data for periods 1 and 2 contain significant gaps without a guarantee that there is no observation in progress. It is also observed that period 3 has the largest number of Srcs (31 sources) compared to periods 1 and 2.

The analysis of the available connection duration is done for three periods namely morning (08:00 - 12:00) shown in epochs 1 to 12, mid-afternoon (12:01-14:05) shown in epochs 13 to 19 and late afternoon (14:06-17:00) in epochs 20 to 29. The result is shown in Figure 4.

Results show increased activity in the morning followed by late afternoon and the in the mid-afternoon. The longest connection duration occurs in the mid afternoon for 80 secs and a minimum of 30 secs occurs twice in the morning, thrice in the mid afternoon and once in the late afternoon.

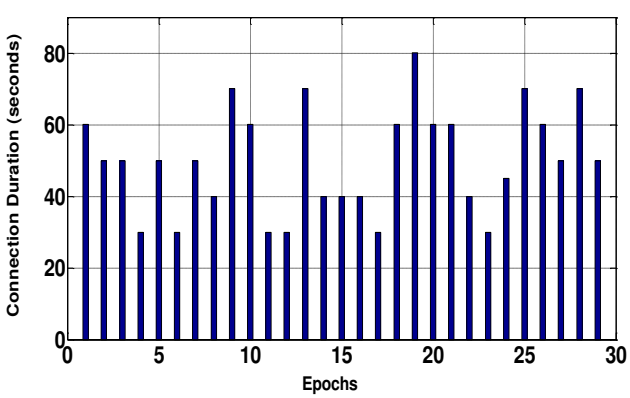

Figure 4: Back to back connection duration.

B. Investigation of End to End Delay

The end to end delay obtained for a given number of satellites in the network (network size) is investigated via MATLAB numerical simulation of (3), (4) and (5) and shown in Figure 5. Equations (3), (4) and (5) describe the end to end delay of the satellite network in the absence of the framework (scenario 1), when the framework operates in a satellite network comprising only ISLs (scenario 2) and both IOLs and ISLs (scenario 3) respectively.

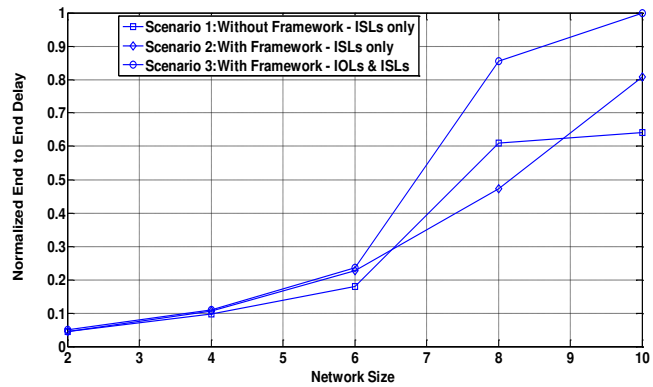

Figure 5: End to End Delay against network size.

From the simulation procedure, it is observed that changing the number of ISLs does not change the trend of the end to end delay with increasing network size. This shows that the sum total of the effects of varying ISL Shannon channel capacity, data size, processing requirement and the network size dominate over the number of ISLs in determining the end to end delay. It can also be seen from Figure 5 that the end to end delay increases with the network size for the three scenarios. This is because the data is forwarded through a larger number of satellites outside the NRQZ vicinity given the fixed number of satellites in the NRQZ vicinity.

Further investigation also shows that the framework increases the end to end delay by 16.7 % and 29.6 % on the average when scenario 1 (without the framework) is compared to scenarios 2 and 3 respectively. Therefore, it can be concluded that the use of IOLs cause a degradation in the end to end delay. This is because of the larger data amounts handled by geostationary satellites with that transmit to more satellites and have a wider coverage.

Analysis shows that introducing the framework increases the end to end delay by 0.5ms and 0.9ms in scenarios 2 and 3 respectively given that the end to end delay in the absence of the framework is 3ms. We also compute the ratio of the increase in the end to end delay in scenarios 2 and 3 compared to scenario 1 for different values of the end to end delay in the range of 3ms to 10ms. This shows that the ratio of the increase in end to end delay obtained in scenario 3 to that of scenario 2 is 1.8. Therefore, scenario 2 is selected because it minimally affects the end to end delay while protecting RASs from interference. Hence, the framework is most suited to low earth orbit satellite networks.

The proposed framework is suitable for non-real time services such as file and video downloads. For example in the presence of the framework, a video download session that

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would have been completed in 40 secs takes an extra 6.7secs. This is a small increment that is not noticeable to the user.

VI. CONCLUSION

This paper identifies that the current knowledge of the universe derivable from radio astronomical observations is impaired because of interfering satellite networks with intersatellite links. We examine the feasibility of using cognitive radio spectrum sharing models for protecting radio astronomical services from this interference. The interweave spectrum sharing model is identified to be suitable because of the advantages of low implementation costs and its ability to improve the accuracy of knowledge gained via astronomical observations. An interference protection framework that deactivates intersatellite links during the epoch of observing cosmic sources is proposed based on this model. Analysis of the spectrum usage of the radio astronomical service shows that the daily spectrum utilization is low and that transmit opportunities exist for constellations of low earth orbiting satellties. The advantages of the proposed framework are that it improves spectrum utilization by allowing the satellites networks to coexist with radio astronomical services, increases the accuracy of knowledge obtained from astronomical observations and does not significantly increase the end to end delay of the satellite network.

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